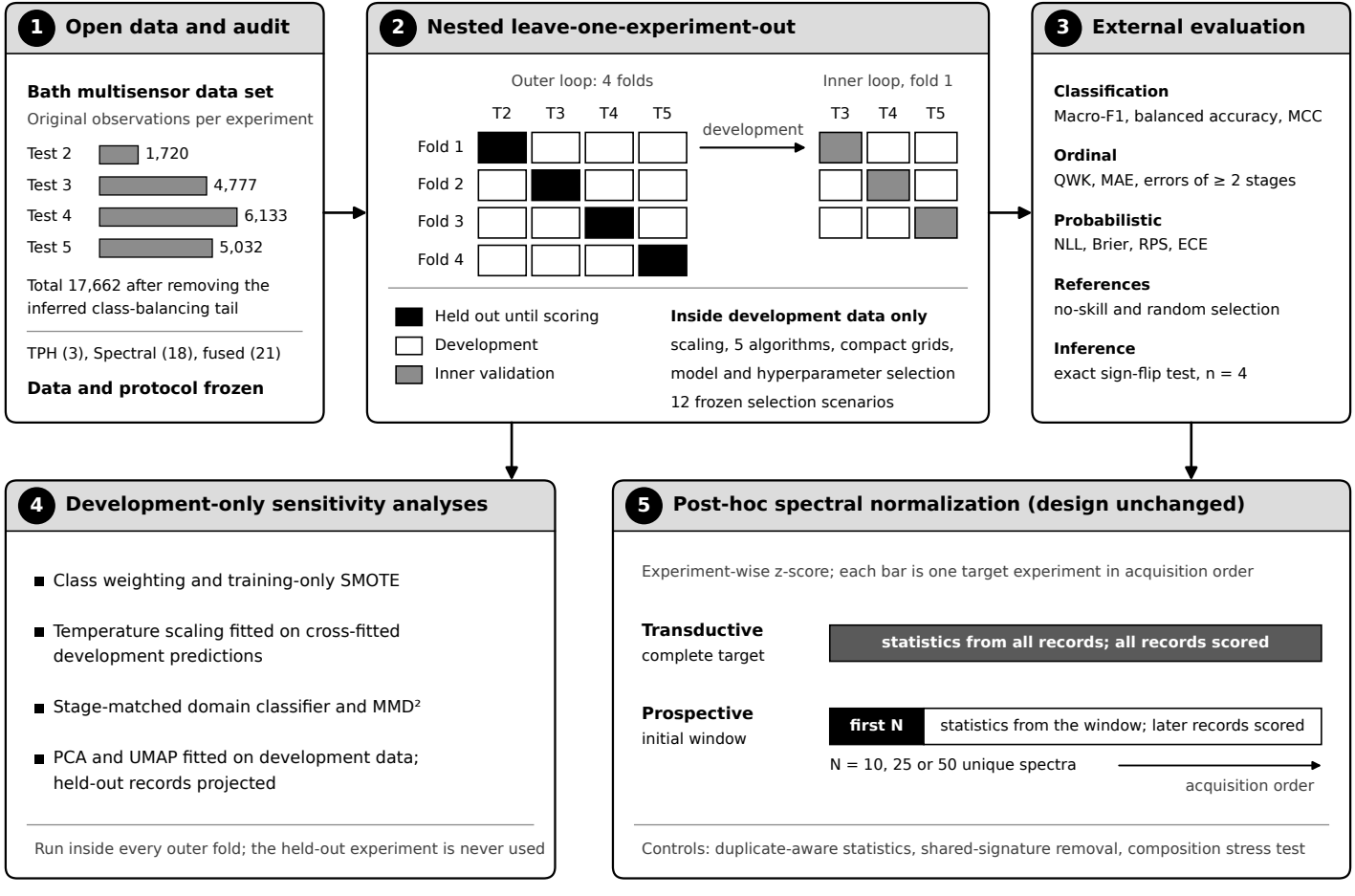
1 Open data and audit
Bath multisensor data set
Original observations per experiment
Test 2 1,720
Test 3 4,777
Test 4 6,133
Test 5 5,032
Total 17,662 after removing the inferred class-balancing tail
TPH (3), Spectral (18), fused (21)
Data and protocol frozen
2 Nested leave-one-experiment-out
Outer loop: 4 folds
T2 T3 T4 T5
Fold 1
Fold 2
Fold 3
Fold 4
development
Inner loop, fold 1
T3 T4 T5
Held out until scoring
Development
Inner validation
Inside development data only
scaling, 5 algorithms, compact grids,
model and hyperparameter selection
12 frozen selection scenarios
3 External evaluation
Classification
Macro-F1, balanced accuracy, MCC
Ordinal
QWK, MAE, errors of ≥ 2 stages
Probabilistic
NLL, Brier, RPS, ECE
References
no-skill and random selection
Inference
exact sign-flip test, n = 4
4 Development-only sensitivity analyses
Class weighting and training-only SMOTE
Temperature scaling fitted on cross-fitted development predictions
Stage-matched domain classifier and MMD²
PCA and UMAP fitted on development data; held-out records projected
Run inside every outer fold; the held-out experiment is never used
5 Post-hoc spectral normalization (design unchanged)
Experiment-wise z-score; each bar is one target experiment in acquisition order
Transductive
complete target
statistics from all records; all records scored
Prospective
initial window
first N
statistics from the window; later records scored
N = 10, 25 or 50 unique spectra
acquisition order
Controls: duplicate-aware statistics, shared-signature removal, composition stress test